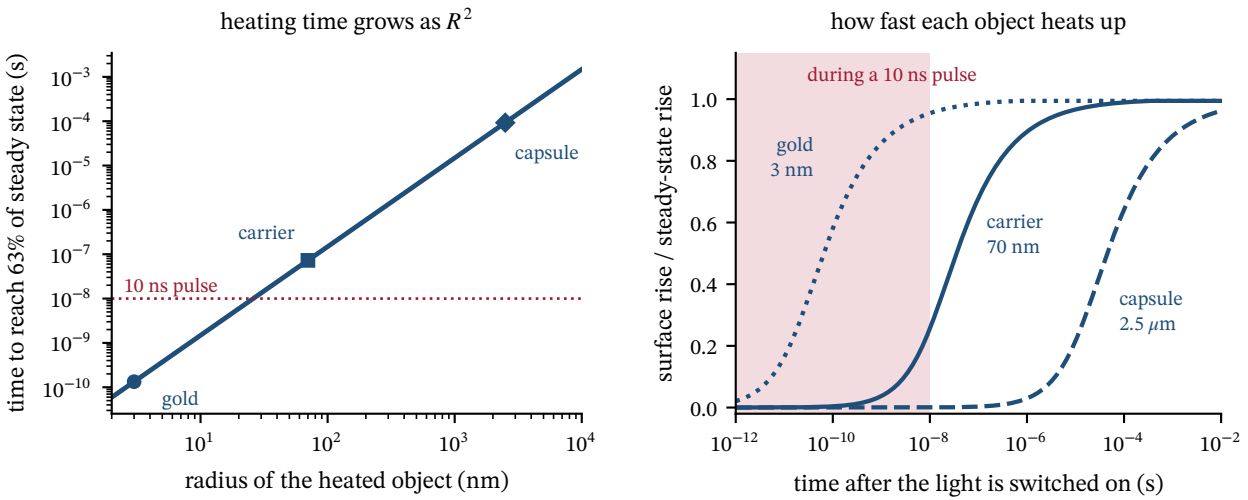
heating time grows as R^2
time to reach 63% of steady state (s)
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
10^{-9}
10^{-10}
capsule
carrier
10 ns pulse
gold
10^{1}
10^{2}
10^{3}
10^{4}
radius of the heated object (nm)
how fast each object heats up
surface rise / steady-state rise
during a 10 ns pulse
gold
3 nm
carrier
70 nm
capsule
2.5 μm
1.0
0.8
0.6
0.4
0.2
0.0
10^{-12}
10^{-10}
10^{-8}
10^{-6}
10^{-4}
10^{-2}
time after the light is switched on (s)